

UTM TENSILE REPORT — UTM_Test_20260817_161046

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

19.9 MPa

UTS

18.3 MPa

Yield σ_y

1.26 GPa

Modulus E

4.2 %

Failure ϵ_f

745 kJ/m³

Toughness

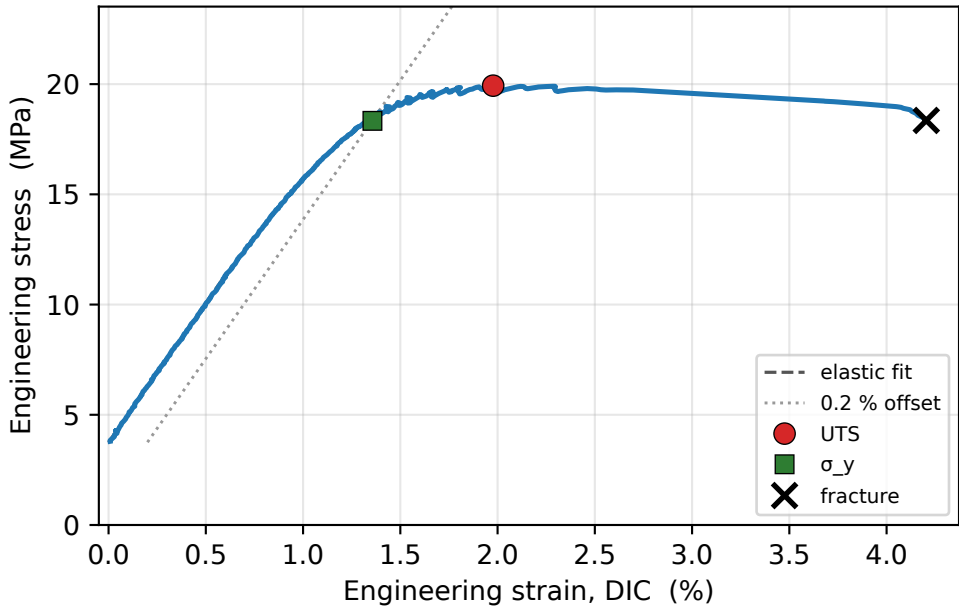
302 N

Preload anchor

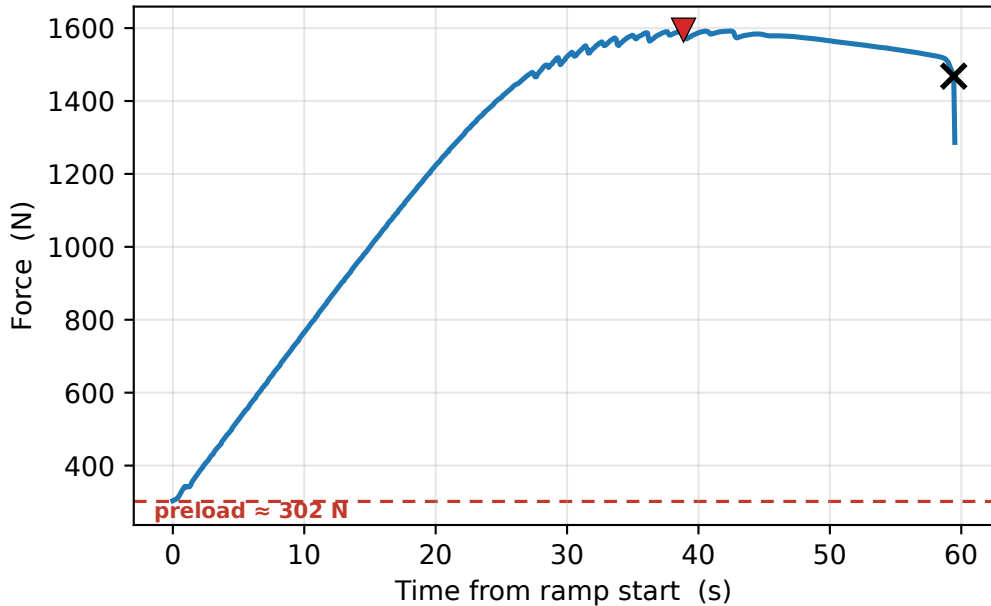
TEST SETTINGS

Specimen	UTM_Test_20260817_161046
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 326.404
Comment	—
Test date	2026-08-17 16:07:52
Duration	179.6 s

Stress-strain



Load vs time

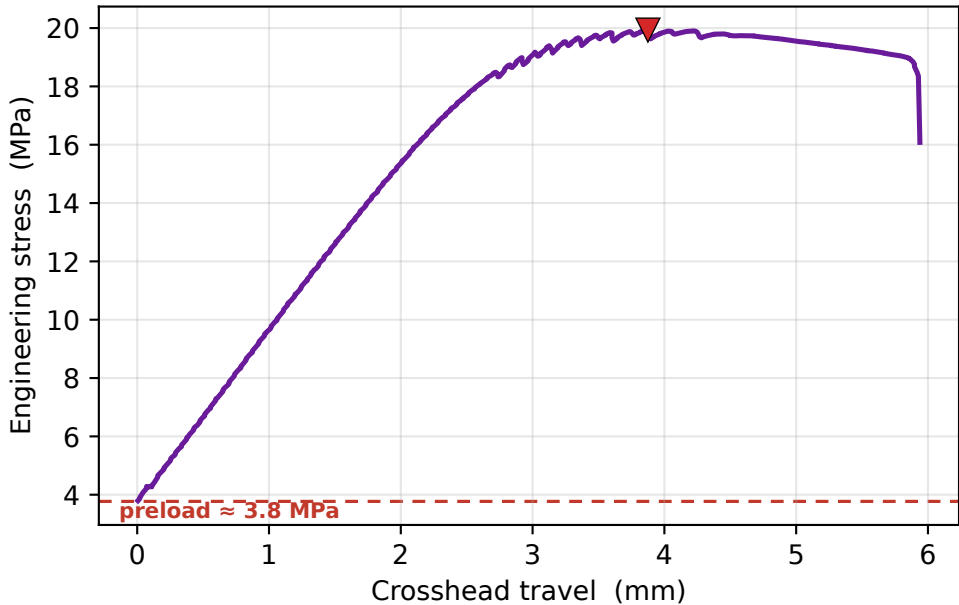


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	19.9	32-60 low	×2.91
σ_y	18.3	30-50 low	×3.16
E	1.3	3-5.5 low	×2.27
ϵ_f	4.2 %	(range 3-7 %)	×1.90

⚠ strength outside journal range

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

